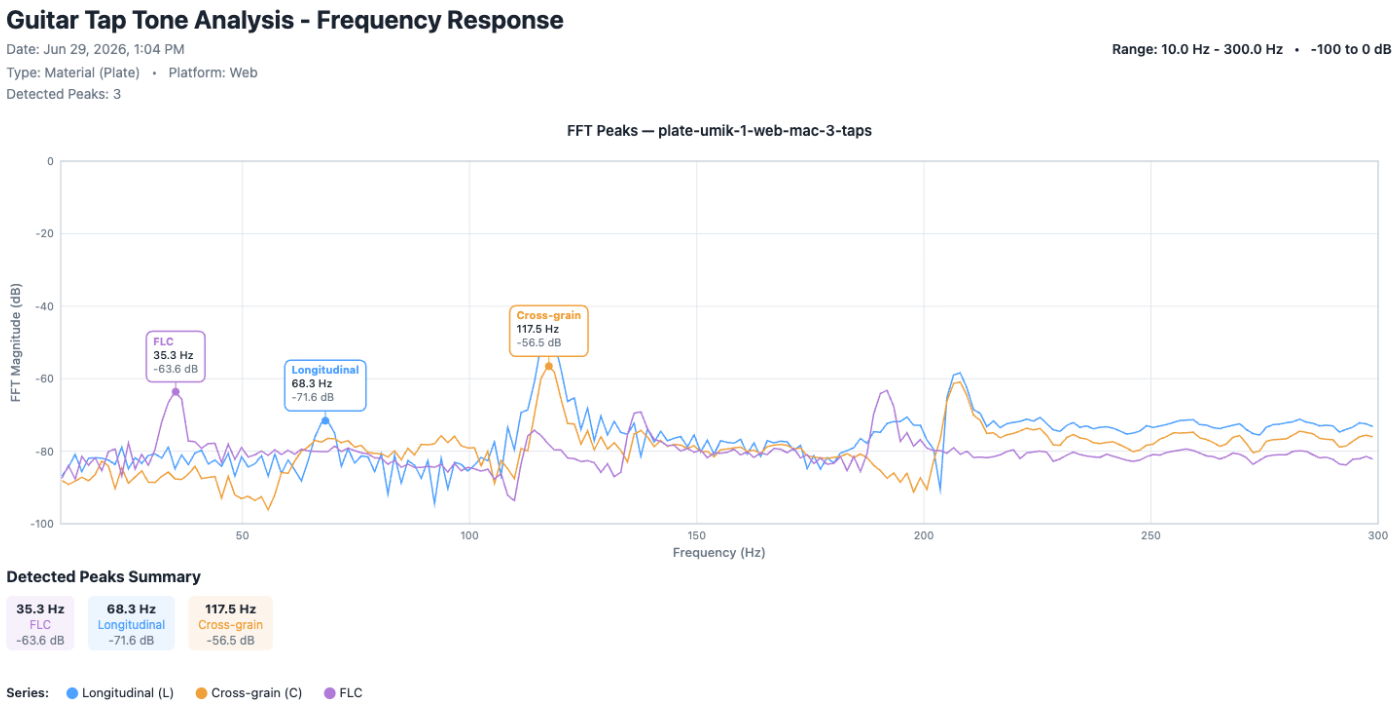
GuitarTap

Tap Tone Analysis Report

Jun 29, 2026, 1:04 PM

Measurement Name:	plate-umik-1-web-mac-3-taps
Type:	Material (Plate)
Notes:	Chrome capture of 3 tap full plate measurement
Frequency Range:	10 Hz – 300 Hz
Microphone:	Umik-1 Gain: 18dB (2752:0007) · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
35.3 Hz	-63.6 dB	C#1	6.0	FLC (Diagonal)
68.3 Hz	-71.6 dB	C#2	15.7	Longitudinal (L)
117.5 Hz	-56.5 dB	A#2	26.7	Cross-grain (C)

Gore Target Thickness

2.83 mm

GLC (Shear Modulus): 0.340 GPa
Body: 490 × 390 mm · f_vs = 60 (Classical Top)

Plate Properties

Sample Dimensions

Length: **557.5 mm**

Width: **220.5 mm**

Thickness: **4.85 mm**
Density: **0.349 g/cm³**

Mass: **208.0 g**

fL: 68.3 Hz fC: 117.5 Hz fLC: 35.3 Hz

Speed of Sound (L): **4256 m/s**
Young's Modulus (L):**6.32 GPa**
Specific Modulus (L): **18.1** *(Fair)*
Radiation Ratio (L): **12.2**

Speed of Sound (C): **1146 m/s**
Young's Modulus (C):**0.46 GPa**
Specific Modulus (C): **1.3** *(Very Good)*
Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.072** *(typical: 0.04–0.08)*

Long/Cross Ratio: **13.8** *(typical: 12–25)*

Overall Quality: Good

Three-Tap Measurement Process:

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.